

David R. Miller

PHD CANDIDATE - UNIVERSITY OF BRITISH COLUMBIA

✉ drmillar@phas.ubc.ca

🏠 davidmillerastro.com

🆔 ORCID: 0000-0002-4591-1903

Education

University of British Columbia

Vancouver, BC, Canada

PHD, ASTRONOMY

2022–Present

Supervisors: Dr. Jeremy Heyl and Dr. Harvey Richer (1944–2023)

Thesis: Probing Stellar Evolution With White Dwarfs in Open Clusters

Expected Completion: Spring 2027

University of British Columbia

Vancouver, BC, Canada

MSc, ASTRONOMY

2020–2022

Supervisor: Dr. Harvey Richer (1944–2023)

Thesis: The White Dwarf Initial-Final Mass Relation: Ultramassive White Dwarfs Ejected from Young Open Star Clusters

Simon Fraser University

Burnaby, BC, Canada

BSc, PHYSICS HONOURS (WITH DISTINCTION)

2016–2020

Supervisor: Dr. Levon Pogorian

Thesis: Dark Sector Cosmic Fluid Interactions as a Solution to the Observed Disagreement in Measurements of the Hubble Expansion Rate

Awards

UBC Four Year Fellowship (4YF), \$98,000

2023–2027

President's Academic Excellence Initiative PhD Award, \$4,800

2022–2026

Dunlap Institute Summer Undergraduate Research Program Fellowship (SURP), \$9,500

2019

NSERC Undergraduate Student Research Award (USRA), \$5,250

2018

Ken Caple College Transfer Entrance Scholarship, \$3,500

2016

Publications

First Author (4)

1. **Miller, D. R.**, Heyl, J., and Tremblay, P-E. Investigating the Observational Progenitor Mass Gap in the White Dwarf Initial-Final Mass Relation. I. Cluster Census and Characterization of the First White Dwarfs in the Gap. 2026, *ApJ*, 1007:183.
2. **Miller, D. R.**, Caiazzo, I., Heyl, J., et al. The White Dwarf Initial-Final Mass Relation from Open Clusters in Gaia DR3. 2026, *ApJ*, 996:69.
3. **Miller, D. R.**, Caiazzo, I., Heyl, J. et al. An Extremely Massive White Dwarf Escaped From the Hyades Star Cluster. 2023, *ApJL*, 956:L41. **Featured in:** AAS Nova, Space.com, Universe Today.
4. **Miller, D. R.**, Caiazzo, I., Heyl, J. et al. The Ultramassive White Dwarfs of the Alpha Persei Cluster. 2022, *ApJL*, 926:L24.

Second/Third Author (3)

1. Jadhav, V. V., Kroupa, P., **Miller, D. R.**, Sahu, S., and Franktisek, D. The white dwarf population of open

clusters and their tidal tails: Tracers of contamination and stellar interactions. 2026, *A&A*, accepted.

2. Yan, H., **Miller, D. R.**, Zhao, J., Guo, J., and Li, C. Escaped White Dwarf Candidates from Open Clusters. 2026, *ApJS*, 286:5.
3. Yan, H., Wang, L., **Miller, D. R.**, et al. Searching for White Dwarf Candidates Formed Through Binary Evolution in Star Clusters. 2026, *ApJ*, 999:167.

Contributing Author (4)

1. Cristea, A. A., Caiazzo, I., Cunningham, T., Raymond, J. C., Vennes, S., Kawka, A., Desai, A., **Miller, D. R.**, et al. A half ring of ionized circumstellar material trapped in the magnetosphere of a white dwarf merger remnant: A new class of white dwarf merger remnants with X-ray emission. 2026, *A&A*, 706, A188.
2. Côté, P., et al. (including **Miller, D. R.**). The CASTOR mission. 2025, *JATIS*, 11(4).
3. Richer, H. B., Caiazzo, I., Du, H., Grondin, S., Hegarty, J., Heyl, J., Kerr, R., **Miller, D. R.**, and Thiele, S. Massive White Dwarfs in Young Star Clusters. 2021, *ApJ*, 912:165. (Author order alphabetical after the first two)
4. Tzanavaris, P., Gallagher, S. C., Ali, S., **Miller, D. R.**, Peninga, S., and Johnson, K. E. Cosmic Pathways for Compact Groups in the Milli-Millennium Simulation. 2019, *ApJ*, 871:242.

Unpublished and In Preparation

1. **Miller, D. R.**, Yan, H., et al. Characterization of Candidate High-Mass White Dwarf Escapees Candidates from Open Clusters. In prep.
2. Yan, H., **Miller, D. R.**, Li, C., et al. Photometric and Spectroscopic Analysis of an Extremely Low-Mass White Dwarf Binary. In prep.
3. **Miller, D. R.**, Caiazzo, I., and Heyl, J. Atypical White Dwarfs in Open Clusters. In prep.
4. **Miller, D. R.**, Heyl, J. A Spectroscopic Census of Cluster White Dwarfs from DESI, LAMOST, and SDSS. In prep.
5. Heyl, J., Caiazzo, I., Richer, H., and **Miller, D. R.**. Reconstructing Nearby Young Clusters with Gaia EDR3. *arXiv pre-prints* arXiv:2110.04296.

Observing Programs as PI

Principle investigator on 15 successful observing programs with Gemini, CFHT, and HST, with a strong record of securing competitive telescope time

Gemini	<i>Spectroscopic confirmation of escaped high-mass white dwarfs for the cluster initial-final mass relation</i>	2026
Gemini	<i>Expanding the high-mass end of the cluster white dwarf initial-final mass relation</i>	2026
HST (Co-PI)	<i>Hidden in Plain Sight? Unraveling the Potential Merger Histories of Ultra-Massive White Dwarfs in the Initial-Final Mass Relation</i>	2026
Gemini	<i>Tracing contamination and stellar interactions in the white dwarf populations of open clusters and their tidal tails</i>	2026
CFHT	<i>Probing the mass gap in the white dwarf initial-final mass relation with the rich intermediate-aged open cluster NGC 2360</i>	2026
Gemini	<i>Probing Accelerated Cooling and Formation in a White Dwarf Binary from a Young Open Cluster</i>	2026

CFHT	<i>Probing the white dwarf initial-final mass relation mass gap in intermediate age open clusters with CFHT MegaCam</i>	2025
Gemini	<i>Probing Non-Monotonic Trends in the Open Cluster White Dwarf Initial-Final Mass Relation</i>	2025
Gemini	<i>Developing the White Dwarf Initial-Final Mass Relation by Advancing the Survey of High-Mass White Dwarfs in Open Star Clusters</i>	2024
Gemini	<i>Improving the Census of High-Mass White Dwarfs in Open Star Clusters</i>	2024
Gemini	<i>High Mass White Dwarf Escapees From Clusters</i>	2023
Gemini	<i>An Extremely Massive White Dwarf Member of the vdB-Hagen 99 Cluster</i>	2022
Gemini	<i>An Ultramassive White Dwarf Escaped From the Hyades Cluster</i>	2021
Gemini	<i>Reconstructing Nearby Young Clusters with Gaia EDR3: Two White Dwarfs Escaped from Alpha Per</i>	2021
Gemini	<i>Spectroscopy of Massive White Dwarfs in Extended Survey Around Young Clusters</i>	2021

Select Talks and Presentations

Klosterneuburg, Austria	24th European White Dwarf Workshop, ISTA	Contributed Talk	2026
Halifax, Canada	CASCA (Canadian Astronomical Society) 2025	Contributed Talk	2025
Burnaby, Canada	Cosmology Seminar, Simon Fraser University	Invited Talk	2025
Barcelona, Spain	23rd European White Dwarf Workshop	Contributed Talk	2024
Padova, Italy	EAS (European Astronomical Society) 2024	e-Poster	2024
Toronto, Canada	CASCA 2024: Harvey Richer Special Memorial Session	Invited Talk	2024
Vancouver, Canada	Astronomy Seminar, University of British Columbia	Invited Talk	2022
Tübingen, Germany	22nd European White Dwarf Workshop	Contributed Talk	2022

Teaching

Note: After 2023, I did not take on additional TA assignments because my graduate fellowship removed the requirement and my supervisor wanted me to prioritize research. In 2026, I was specifically asked to TA the biennial graduate Stellar Astronomy course, reflecting my standing as the department's leading graduate student in stellar astrophysics.

Astro 508: Stellar Astronomy , TA	2026
Sole TA for this graduate course. Duties include holding office hours, supporting student learning, and managing grading.	
Astro 310: The Solar System , Head Tutorial TA	2022–2023
Oversaw tutorials, managed the course gradebook, held office hours, and revised tutorial materials now used in subsequent offerings.	
Astro 311: Stars and Galaxies , Tutorial TA	2023
Astro 102: Intro to Stars and Galaxies , Tutorial TA	2022
Phys 131: Energy and Waves , Tutorial TA	2021
Astro 101: Introduction to the Solar System , Tutorial TA	2020–2021
Ran and supported tutorials, held office hours, and marked coursework.	

Academic Service & Leadership

JOURNAL REFEREE – <i>ApJ</i>	2026–present
VOLUNTEER – Testing Gravity, Vancouver, BC	2019, 2025

SOLE ORGANIZER – Compact Object Journal Club, <i>University of British Columbia</i>	2021–2024
PROBLEM SETTER AND JUDGE – High-Energy Astrophysics Olympiad, <i>University of British Columbia</i>	2022

Outreach

PUBLIC TALKS

Space Camp Saturdays, <i>H. R. MacMillan Space Centre, Vancouver, BC</i>	2026
Science Rendezvous, <i>University of British Columbia</i>	2026

PUBLIC ENGAGEMENT

Astronomy on Tap, <i>University of British Columbia</i>	2024–present
Astronomy on Tap, <i>University of Toronto</i>	2019
Cronyn Observatory, <i>University of Western Ontario</i>	2018

Research Affiliations & Professional Development

Legacy Survey of Space and Time (LSST), <i>Canada</i>	2025–present
Provisional international data rights holder through Canada’s in-kind contribution to the Rubin Observatory.	
Summer School in Statistics for Astronomers, <i>Penn State</i>	2026
Training in statistical inference, regression and model selection, resampling, Bayesian analysis, MCMC and nested sampling, time series analysis, spatial statistics, and machine learning in R.	
ML4Astro Workshop – ICML 2025, <i>Vancouver, BC</i>	2025
Focused on modern machine-learning methods in astrophysics, including foundation models, simulation-based inference, anomaly detection, and AI-assisted scientific workflows.	
Testing Gravity Pre-Conference School, <i>Vancouver, BC</i>	2025
Advanced review of gravitation and cosmology in preparation for the Testing Gravity conference.	
CRAQ Summer School on <i>Space Astronomy</i> , <i>Montreal, QC</i>	2023
Training in space-based astronomy, satellite operations, and the development of astronomical missions and instrumentation.	
CASTOR Science Team - Stellar Astrophysics Working Group, <i>Canada</i>	2022–2023
Contributed to development of the stellar astrophysics science case for CASTOR, a proposed Canadian-led UV-optical space telescope.	
Testing Gravity Pre-Conference School, <i>Vancouver, BC</i>	2019
Introductory training in gravitation and cosmology in preparation for the Testing Gravity conference.	
CRAQ Summer School on <i>Large-Scale Astrophysics</i> , <i>Montreal, QC</i>	2018
Training in galaxy formation and evolution, galaxy clusters, and the use of large-scale structure as a cosmological probe.	